# Minxing Zhang

Fourth-Year Ph.D. Candidate in Computer Science | Duke University
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**Research focus.** My research centers on privacy, generalizability, and reliable, cost-efficient evaluation for trustworthy LLMs and AI agents. My recent work explores reliable person distillation and representation, advertising for LLM agents, and question-asking capabilities in LLMs.

## Education

**Duke University** 2023–Present
Ph.D. in Computer Science *(Expected Graduation: December 2027)*
Durham, NC | Adviser: Jian Pei

**Emory University** 2019–2022
B.S. in Computer Science and Applied Mathematics & Statistics
Atlanta, GA | GPA: 3.984/4.00; Highest Honor | Advisers: Li Xiong and Liang Zhao

## Work and Internship Experience

### Amazon Web Services | Applied Scientist Intern
Summer 2026
Amazon Bedrock Guardrails, Deep Engine-Science | Seattle, WA
*Research paper in submission*

Developed **Judge2Vec**, an unsupervised framework that selects compact, complementary LLM juries using criterion-specific competence embeddings estimated from leave-one-family-out agreement with a family-balanced consensus on unlabeled pilot examples.

Selected **4 of 19 judges**, ranking first among comprehensive baselines on 8 text benchmarks. Kendall correlation reached 0.467 vs. 0.424 for full-pool aggregation with **83.9% lower deployment cost**.

Enabled zero-shot jury transfer without target-task examples, retaining 98.5% of performance; added budget adaptation and extended evaluation to multimodal and agentic settings.

### AT&T Chief Data Office | Graduate Research Intern
Summer 2025
Bedminster, NJ | Mentors: Divesh Srivastava and Yaron Kanza
*Research paper in submission*

Developed a kNN-LM framework for question answering and **selective forgetting without retraining the LLM or embedding function**. Designed embedding functions that preserve accurate next-token retrieval while protecting against leakage of deleted private associations: sensitive values are masked during embedding training, while correct next tokens remain retrievable from the private database until deletion.

Achieved **73.8% answer accuracy vs. 60.0%** for RAG.

Implemented FAISS retrieval and user-level logical deletion without index rebuilds; scaling the synthetic datastore 1,000-fold to **19.6 million key–value pairs** increased generation time only 4.9-fold (1.42–7.01 seconds/query).

### Microsoft & Emory University | DataThink Research Project
Summer 2022
Atlanta, GA

Built XGBoost rental-price prediction and SKATER spatial clustering models to analyze metro Atlanta's rental market.

## Selected Research Projects

**Person distillation.** Formulated person distillation as modeling individual knowledge, judgment, and behavior from heterogeneous personal traces; established a lifecycle taxonomy and framework for evaluating person-specific fidelity.

**LLM-based agent generalizability.** Defined agent generalizability and organized evaluation and improvement methods by agent component; distinguished generalizable frameworks from agents that generalize to unseen settings.

**MPCEval.** Developed a reference-free, reproducible benchmark for multi-party conversation generation, with turn- and conversation-level metrics for speaker modeling, content quality, and speaker–content consistency. Benchmarked diverse generation methods across datasets spanning collaborative problem-solving, scripted dialogue, and real-world enterprise discussions. **KDD 2026 Datasets and Benchmarks Track, oral.**

**Privacy attacks and defenses.** Studied membership-inference attacks on teacher–student pairs and proposed mitigations for privacy leakage in knowledge distillation. Also developed a temporal attack on sequential data releases using hidden Markov models and reinforcement learning.

**Earlier research at Emory.** Three first-author papers on spatial interpolation, self-supervision, and trajectory generation.

## Technical Skills

**Languages:** Python, Java, R, SQL | **Research tools:** LaTeX

## Publications

### Published and Accepted

**[1] MPCEval: A Benchmark for Multi-Party Conversation Generation**
**Minxing Zhang**[*], Yi Yang[*], Zhuofan Jia, Xuan Yang, Jian Pei, Yuchen Zang, Xingwang Deng, Xianglong Chen
**ACM SIGKDD**, Datasets and Benchmarks Track; **oral presentation**. [Paper] [Code] 2026

**[2] Generalizability of Large Language Model-Based Agents: A Comprehensive Survey**
**Minxing Zhang**, Yi Yang, Roy Xie, Bhuwan Dhingra, Shuyan Zhou, Jian Pei
**ACM Computing Surveys**, 58(10), Article 263, 44 pages. [Paper] 2026

**[3] On Membership Inference Attacks in Knowledge Distillation**
Ziyao Cui[*], **Minxing Zhang**[*], Jian Pei
**ACM SIGKDD Explorations Newsletter**, 28(1), pp. 32–40. [Paper] 2026

**[4] Protecting Data Buyer Privacy in Data Markets**
**Minxing Zhang**, Jian Pei
**IEEE Internet Computing**, 28(4), pp. 14–20. [Paper] 2024

**[5] CSGAN: Modality-Aware Trajectory Generation via Clustering-Based Sequence GAN**
**Minxing Zhang**, Haowen Lin, Shun Takagi, Yang Cao, Cyrus Shahabi, Li Xiong
**IEEE MDM**, pp. 148–157. [Paper] 2023

**[6] Deep Spatial Prediction via Heterogeneous Multi-Source Self-Supervision**
**Minxing Zhang**, Dazhou Yu, Yun Li, Liang Zhao
**ACM Transactions on Spatial Algorithms and Systems**, 9(3). [Paper] 2023

**[7] Deep Geometric Neural Network for Spatial Interpolation**
**Minxing Zhang**, Dazhou Yu, Yun Li, Liang Zhao
**ACM SIGSPATIAL**; Best Poster Runner-Up. [Paper] 2022

### Preprints

**[8] Person Distillation: Foundations and Principles for Personalized AI**
**Minxing Zhang**[*], Yi Yang[*], Jian Pei
Preprints.org, 202608.1530.v1. [Paper] [Website] [Github] 2026

**[9] Learning to Attack: Uncovering Privacy Risks in Sequential Data Releases**
Ziyao Cui[*], **Minxing Zhang**[*], Jian Pei
arXiv:2510.24807. [Paper] 2025

## Tutorials and Invited Talks

**KDD | Tutorial Co-organizer** 2026
Trustworthy LLM-Based Agents for Data-Centric AI. [Website]

**PAKDD | Tutorial Co-organizer** 2026
Generalizability of Large Language Model-Based Agents. [Program]

**Samsung ML Reading Group | Invited Speaker** 2024

## Honors and Awards

**Academic Excellence Award,** Emory Computer Science 2023
**Phi Beta Kappa,** Gamma Chapter of Georgia 2023
**Best Poster Runner-Up,** ACM SIGSPATIAL 2022
**Travel Grant,** ACM SIGSPATIAL 2022